Boolean Minimization

Lecture 4

Outline

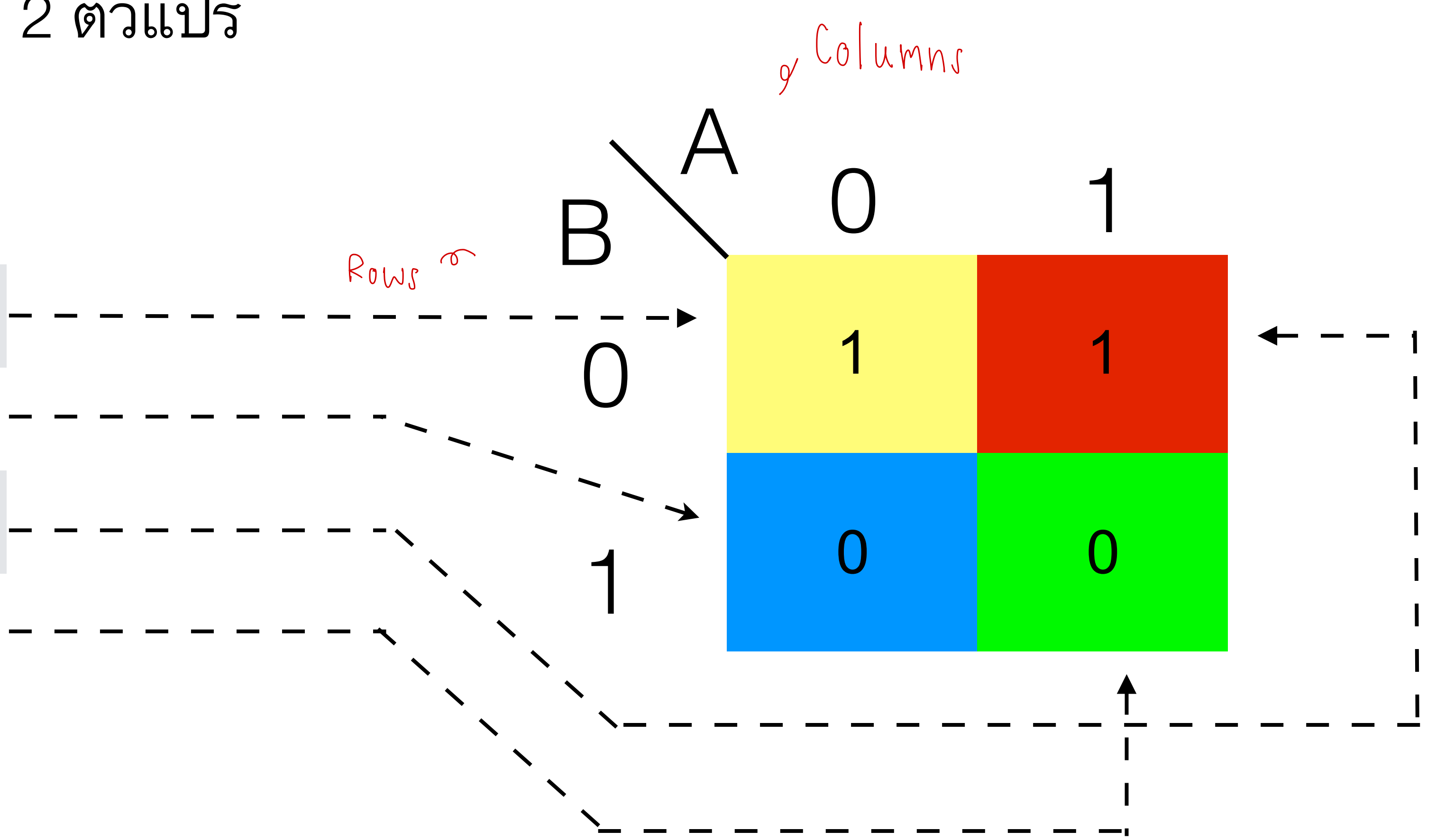
- Karnaugh-map
- Circuit design

Karnaugh Map

- Karnaugh Map (K-map) อาศัยการจัดเรียงตารางค่าความจริงในรูปแบบใหม่ ซึ่งทำให้สามารถลดรูปโดยอาศัยคุณสมบัติการคอมพลีเมนต์ได้ง่ายขึ้น

- ตัวอย่างของ K-map แบบ 2 ตัวแปร

A	B	F
0	0	1
0	1	0
1	0	1
1	1	0



ลดรูปสมการ

$\boxed{F = \bar{B}}$

Examples of K-Map

Colums

Rows

C \ AB	00	01	11	10
0				
1				

ABC

3-variable K-map

CD \ AB	00	01	11	10
00				
01				
11				
10				

ABCD

4-variable K-map

Using K-Map

- หาช่องใน K-map ที่มีลอจิก 1 ติดกัน
- จำนวนช่องที่ใช้ลดรูปต้องมีขนาด 2, 4, 8, ... เท่านั้น
- ตัวอย่าง

A	B	F
0	0	1
0	1	0
1	0	1
1	1	0

B \ A	0	1
0	1	1
1	0	0

$$F = \overline{B}$$

Using K-Map (2)

- ตัวอย่าง

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

C \ AB	00	01	11	10
0	0	0	1	1
1	0	1	1	1

Output จะเป็น 1 เมื่อ A=1 หรือ B=1 และ C=1

$$F = A + BC$$

SOP

Example

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

C \ AB	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$$F = AB + BC + AC$$

Example (2)

A	B	C	F
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

C \ AB	00	01	11	10
0	1	0	0	1
1	0	0	1	1

$$F = AC + \bar{B}\bar{C}$$

Example (3)

A	B	C	D	F
0	0	0	0	1
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

CD \ AB	00	01	11	10
00	1	0	0	1
01	1	1	1	1
11	1	0	0	0
10	0	0	0	0

$$F = \bar{C}D + \bar{B}\bar{C} + \bar{A}\bar{B}C$$

Example (4)

A	B	C	D	F
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

CD \ AB	00	01	11	10
00	1	0	0	1
01	0	1	0	0
11	1	1	1	1
10	1	1	1	1

$$F = C + \bar{B}\bar{D} + \bar{A}BD$$

Don't Cares

- การใช้ประโยชน์จาก don't cares

- ตัวอย่าง

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	x
1	1	1	x

don't cares

C \ AB	00	01	11	10
0	0	0	x	1
1	0	1	x	0

$$F = BC + A\overline{C}$$

Example (5)

A	B	C	F
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	x
1	0	0	x
1	0	1	1
1	1	0	0
1	1	1	1

C \ AB	00	01	11	10
0	1	0	0	X
1	0	X	1	1

$$F = AC + \bar{B}\bar{C}$$

Example (6)

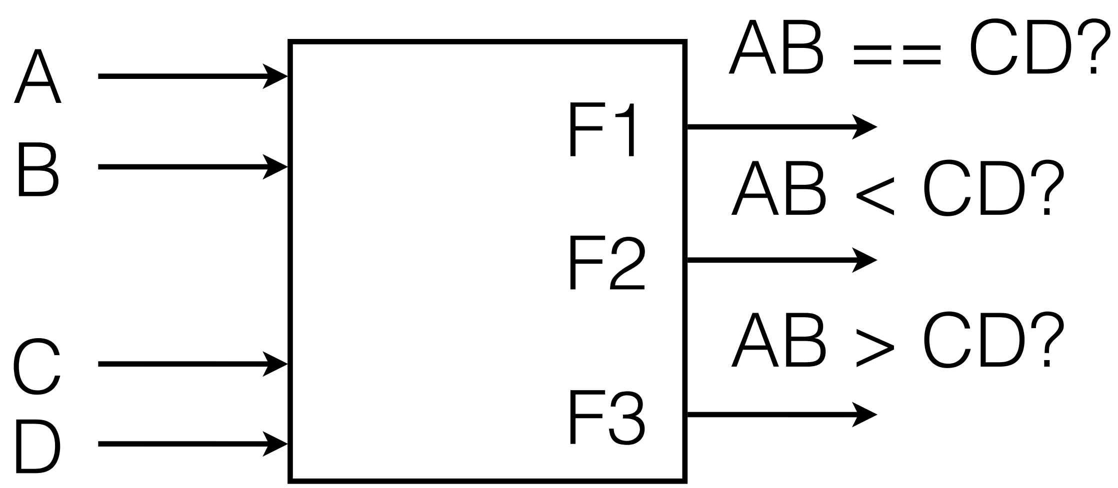
A	B	C	D	F
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	x
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	x
1	1	0	1	x
1	1	1	0	x
1	1	1	1	x

CD \ AB	00	01	11	10
00	1	X	X	1
01	1	1	X	0
11	0	1	X	1
10	1	1	X	0

$$F = \bar{A}\bar{C} + \bar{C}\bar{D} + \bar{A}\bar{D} + B + BC\bar{D}$$

Circuit Design

- ตัวอย่าง: ออกแบบ Two-bit Comparator
- Input Spec:
  - 2-bit input จำนวน 2 ตัว (AB, CD)
- Output Spec:
  - F1 เช็คว่า AB == CD
  - F2 เช็คว่า AB < CD
  - F3 เช็คว่า AB > CD



Two-bit Comparator

AB == CD ?

A	B	C	D	F1	F2	F3
0	0	0	0	1		
0	0	0	1	0		
0	0	1	0	0		
0	0	1	1	0		
0	1	0	0	0		
0	1	0	1	1		
0	1	1	0	0		
0	1	1	1	0		
1	0	0	0	0		
1	0	0	1	0		
1	0	1	0	1		
1	0	1	1	0		
1	1	0	0	0		
1	1	0	1	0		
1	1	1	0	0		
1	1	1	1	1		

CD \ AB	00	01	11	10
00	1	0	0	0
01	0	1	0	0
11	0	0	1	0
10	0	0	0	1

$$F_1 = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}D + ABCD + A\bar{B}C\bar{D}$$

Two-bit Comparator (2)

AB < CD?

A	B	C	D	F1	F2	F3
0	0	0	0		0	
0	0	0	1		1	
0	0	1	0		1	
0	0	1	1		1	
0	1	0	0		0	
0	1	0	1		0	
0	1	1	0		1	
0	1	1	1		1	
1	0	0	0		0	
1	0	0	1		0	
1	0	1	0		0	
1	0	1	1		1	
1	1	0	0		0	
1	1	0	1		0	
1	1	1	0		0	
1	1	1	1		0	

CD \ AB	00	01	11	10
00	0	0	0	0
01	1	0	0	0
11	1	1	0	1
10	1	1	0	0

$$F_2 = \bar{A}C + \bar{A}\bar{B}D + \bar{B}CD$$

Two-bit Comparator (3)

← AB > CD?

A	B	C	D	F1	F2	F3
0	0	0	0			
0	0	0	1			
0	0	1	0			
0	0	1	1			
0	1	0	0			
0	1	0	1			
0	1	1	0			
0	1	1	1			
1	0	0	0			
1	0	0	1			
1	0	1	0			
1	0	1	1			
1	1	0	0			
1	1	0	1			
1	1	1	0			
1	1	1	1			

CD \ AB	00	01	11	10
00				
01				
11				
10				

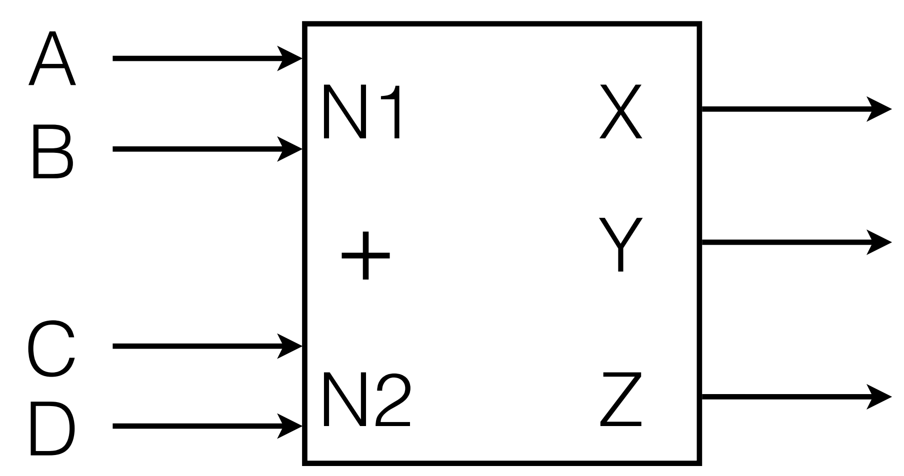
$$F_3 \quad =$$

Two-Bit Comparator (4)

- Schematic Diagram

Two-Bit Binary Adder

- ตัวอย่าง ออกแบบ Two-bit Binary Adder
- Input Spec
  - 2-bit input จำนวน 2 ตัว (AB, CD)
- Output Spec
  - 3-bit binary (XYZ)



Two-bit Binary Adder (2)

A	B	C	D	X	Y	Z
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	0	0	1	0
0	0	1	1	0	1	1
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	0	1	1
0	1	1	1	1	0	0
1	0	0	0	0	1	0
1	0	0	1	0	1	1
1	0	1	0	1	0	0
1	0	1	1	1	0	1
1	1	0	0	0	1	1
1	1	0	1	1	0	0
1	1	1	0	1	0	1
1	1	1	1	1	1	0

CD \ AB	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	0	1	1	1
10	0	0	1	1

$X = AC + BCD + ABD$

Two-bit Binary Adder (3)

A	B	C	D	X	Y	Z
0	0	0	0			
0	0	0	1			
0	0	1	0			
0	0	1	1			
0	1	0	0			
0	1	0	1			
0	1	1	0			
0	1	1	1			
1	0	0	0			
1	0	0	1			
1	0	1	0			
1	0	1	1			
1	1	0	0			
1	1	0	1			
1	1	1	0			
1	1	1	1			

CD \ AB	00	01	11	10
00				
01				
11				
10				

$$Y =$$

Two-bit Binary Adder (4)

A	B	C	D	X	Y	Z
0	0	0	0			
0	0	0	1			
0	0	1	0			
0	0	1	1			
0	1	0	0			
0	1	0	1			
0	1	1	0			
0	1	1	1			
1	0	0	0			
1	0	0	1			
1	0	1	0			
1	0	1	1			
1	1	0	0			
1	1	0	1			
1	1	1	0			
1	1	1	1			

CD \ AB	00	01	11	10
00				
01				
11				
10				

$Z \quad =$

5-Variable K-map

A	B	C	D	E	F	G
0	0	0	0	0	0	1
0	0	0	0	1	1	0
0	0	0	1	0	1	0
0	0	0	1	1	0	0
0	0	1	0	0	1	x
0	0	1	0	1	1	x
0	0	1	1	0	1	1
0	0	1	1	1	0	1
0	1	0	0	0	0	1
0	1	0	0	1	1	0
0	1	0	1	0	1	1
0	1	0	1	1	1	1
0	1	1	0	0	0	x
0	1	1	0	1	1	1
0	1	1	1	0	1	x
0	1	1	1	1	1	x

A	B	C	D	E	F	G
1	0	0	0	0	0	1
1	0	0	0	1	1	0
1	0	0	1	0	0	0
1	0	0	1	1	0	1
1	0	1	0	0	0	1
1	0	1	0	1	1	1
1	0	1	1	0	0	1
1	0	1	1	1	0	1
1	1	0	0	0	0	1
1	1	0	0	1	1	1
1	1	0	1	0	1	1
1	1	0	1	1	1	1
1	1	1	0	0	0	x
1	1	1	0	1	1	1
1	1	1	1	0	0	1
1	1	1	1	1	1	1

ใช้ A ตอน 2 ตาราง
ไม่เชื่อมกัน

A = 0

DE \ BC	00	01	11	10
00	0	1	0	0
01	1	1	1	1
11	0	0	1	1
10	1	1	1	1

A = 1

DE \ BC	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	0	0	1	1
10	0	0	0	1

ติดกัน

$F = BE + \bar{D}E + \bar{A}DE + B\bar{C}D + \bar{A}\bar{B}C\bar{D}$

5-Variable K-map (2)

A	B	C	D	E	F	G
0	0	0	0	0	0	1
0	0	0	0	1	1	0
0	0	0	1	0	1	0
0	0	0	1	1	0	0
0	0	1	0	0	1	x
0	0	1	0	1	1	x
0	0	1	1	0	1	1
0	0	1	1	1	0	1
0	1	0	0	0	0	1
0	1	0	0	1	1	0
0	1	0	1	0	1	1
0	1	0	1	1	1	1
0	1	1	0	0	0	x
0	1	1	0	1	1	1
0	1	1	1	0	1	x
0	1	1	1	1	1	x

A	B	C	D	E	F	G
1	0	0	0	0	0	1
1	0	0	0	1	1	0
1	0	0	1	0	0	0
1	0	0	1	1	0	1
1	0	1	0	0	0	1
1	0	1	0	1	1	1
1	0	1	1	0	0	1
1	0	1	1	1	0	1
1	1	0	0	0	0	1
1	1	0	0	1	1	1
1	1	0	1	0	1	1
1	1	0	1	1	1	1
1	1	1	0	0	0	x
1	1	1	0	1	1	1
1	1	1	1	0	0	1
1	1	1	1	1	1	1

A = 0

DE \ BC	00	01	11	10
00	1	x	x	1
01	0	x	1	0
11	0	1	x	1
10	0	1	x	1

A = 1

DE \ BC	00	01	11	10
00	1	1	x	1
01	0	1	1	1
11	0	1	1	1
10	1	1	1	1

$G = C + AB + \bar{D}\bar{E} + BD + A\bar{E}$

5-Variable K-map (2)

A	B	C	D	E	F	G
0	0	0	0	0	0	1
0	0	0	0	1	1	0
0	0	0	1	0	1	0
0	0	0	1	1	0	0
0	0	1	0	0	1	x
0	0	1	0	1	1	x
0	0	1	1	0	1	1
0	0	1	1	1	0	1
0	1	0	0	0	0	1
0	1	0	0	1	1	0
0	1	0	1	0	1	1
0	1	0	1	1	1	1
0	1	1	0	0	0	x
0	1	1	0	1	1	1
0	1	1	1	0	1	x
0	1	1	1	1	1	x

A	B	C	D	E	F	G
1	0	0	0	0	0	1
1	0	0	0	1	1	0
1	0	0	1	0	0	0
1	0	0	1	1	0	1
1	0	1	0	0	0	1
1	0	1	0	1	1	1
1	0	1	1	0	0	1
1	0	1	1	1	0	1
1	1	0	0	0	0	1
1	1	0	0	1	1	1
1	1	0	1	0	1	1
1	1	0	1	1	1	1
1	1	1	0	0	0	x
1	1	1	0	1	1	1
1	1	1	1	0	0	1
1	1	1	1	1	1	1

A = 0

DE \ BC	00	01	11	10
00	1	x	x	1
01	0	x	1	0
11	0	1	x	1
10	0	1	x	1

A = 1

DE \ BC	00	01	11	10
00	1	1	x	1
01	0	1	1	1
11	0	1	1	1
10	1	1	1	1